

224 - Basic Calculator

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18
(1 + (4 + 5 + 2) - 3) + (6 + 8)

curr_num: int 0

res: int 0

stack: list[int] []

prev_sign = 1 | -1 1

Iterate through the string left → right

- (3 + 4)

-	(	3	+	4
sign = -1	stack = [0, -1]	curr = 3	res = 3	curr = 4
	curr = 0		curr = 0	
	sign = 1		sign = 1	

)

↙ sign of (3 + 4)
[0, -1]

res += curr * sign

res -= stack.pop()

res += stack.pop()

curr = 0

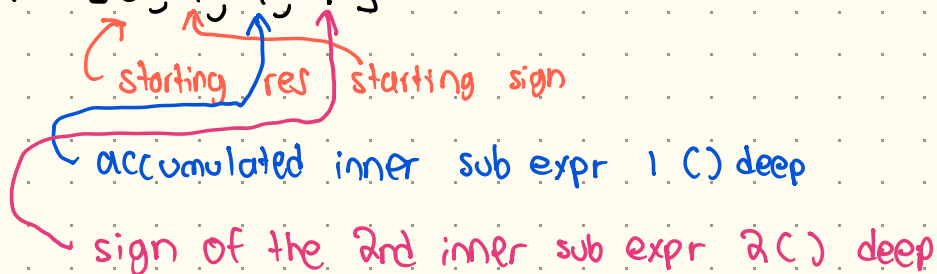
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18
(1 + (4 + 5 + 2) - 3) + (6 + 8)

(stack = [0, 1]

1 curr = 1

+ res = 1 sign = 1

(stack = [0, 1, 1, 1]



4 curr = 4

+ res = 4

curr = 0

sign = 1

5 curr = 5

+ res = 9

curr = 0

sign = 1

2 curr = 2

) res += curr * sign

res = stack.pop()

res += stack.pop()

curr = 0

c.isnumeric() -> build number

' ' -> continue

' (' -> recurse down a sub expression

stack.append(res)

stack.append(sign)

res = 0

sign = 1

)' -> res += sign * curr } Return to

res = stack.pop()

res += stack.pop()

curr = 0

+' -> res += sign * curr

sign = 1

-' -> res += sign * curr

sign = -1